
Handwriting Author Identification



Zainab Siddiqui
Data Capstone II



Preprocessing

- ◆ **Standardizing image dimensions**
 - Adding/removing whitespace
- ◆ **Pre-downsizing**

Standardized Images

The early bird may get the worm, but
the second mouse gets the cheese.

Our London business is good, but Vienna
and Berlin are quiet. Mr. D. Lloyd has gone
to Switzerland and I hope for good news.
He will be there for a week at 1496 Zermatt
Street and then goes to Turin and Rome and
will join Colonel Parry and arrive at Athens,
Greece, November 27 or December 2. Letters
there should be addressed King James Blvd. 3580.
We except Charles E. Fuller Tuesday, Dr. L.
McQuaid and Robert Ungew, Esq. left on the
'Y. K. Express tonight.

Within a short time she was walking briskly
toward the Emerald City, her silver shoes tinkling
merrily on the hard, yellow roadbed. The sun shone
bright and the birds sang sweet and Dorothy
did not feel nearly as bad as you might think a
little girl would who had been suddenly whisked
away from her own country and set down in
the midst of a strange land.

B&W Standardized Images

Our London business is good, but Vienna and Berlin are quiet. Mr D Lloyd has gone to Switzerland and I hope for good news. He will be there for a week at 1496 Zermatt Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Fuller Tuesdays. Dr. L MacQuaid and Robert Unger, Esq., left on the 'Y.X.' Express bright.

The early bird may get the worm, but the second mouse gets the cheese.

Within a short time she was walking briskly toward the Emerald City, her silver shoes twinkling merrily on the hard, yellow roadbed. The sun shone bright and the birds sang sweet and Dorothy did not feel nearly so bad as you might think a little girl would who had been suddenly whisked away from her own country and set down in the midst of a strange land.



Final Model Candidates

-
-
-

Metrics

	Model #2	Model #3	Model #4	Model #5
Train Accuracy	88%	91%	95%	82%
Val. Top-3 Acc.	83%	86%	90%	74%
Test Accuracy	67%	71%	77%	56%
Test Top-3 Acc.	82%	86%	90%	73%



Explainable AI

Model #5

True: w0339
1. w0339 (94.18%) [CORRECT]
2. w0064 (1.71%)
3. w0147 (0.63%)

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope for good news. He will be there for a week at 1496 Tarnett Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Fuller Tuesday. Dr. L. McQuaid and Robert Unger, Esq., left on the 'Y. X.' Express tonight.

True: w0624
1. w0624 (86.65%) [CORRECT]
2. w0547 (4.22%)
3. w0712 (1.68%)

Our London business is good, but Vienna & Berlin are quiet. Mr. D. Lloyd has gone to Switzerland & I hope for good news. He will be there for a week at 1496 Tarnett Street & then goes to Turin & Rome & will join Colonel Perry & arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Fuller Tuesday. Dr. L. McQuaid & Robert Unger, Esq., left on the 'Y. X.' Express tonight.

True: w0495
1. w0234 (7.98%)
2. w0382 (6.39%)
3. w0495 (5.60%) [CORRECT]

THE NEWS DID NOT GET THE NEWS, VIA THE REPORT WERE GET THE NEWS.

True: w0249
1. w0151 (39.56%)
2. w0011 (21.18%)
3. w0348 (7.96%)

There is a lot of news that has been written about the English City, but there are many more things that are not so well known. The sun is bright and the birds are singing and the air is fresh. The sun is bright and the birds are singing and the air is fresh. The sun is bright and the birds are singing and the air is fresh.

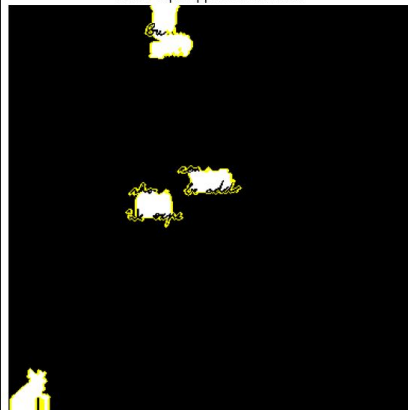
Original
True: w0149

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope for good news. He will be there for a week at 1496 Tarnett Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Fuller Tuesday. Dr. L. McQuaid and Robert Unger, Esq., left on the 'Y. X.' Express tonight.

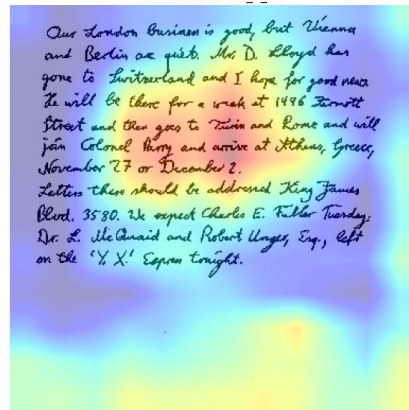
LIME: Pros (Green) / Cons (Red)
LIME top: w0149 | Model: w0149 (81.8%)

Our London business is good, but Vienna and Berlin are quiet. Mr. D. Lloyd has gone to Switzerland and I hope for good news. He will be there for a week at 1496 Tarnett Street and then goes to Turin and Rome and will join Colonel Perry and arrive at Athens, Greece, November 27 or December 2. Letters there should be addressed King James Blvd. 3580. We expect Charles E. Fuller Tuesday. Dr. L. McQuaid and Robert Unger, Esq., left on the 'Y. X.' Express tonight.

LIME: Top Supportive Features



Grad-CAM: Conv_1_bn



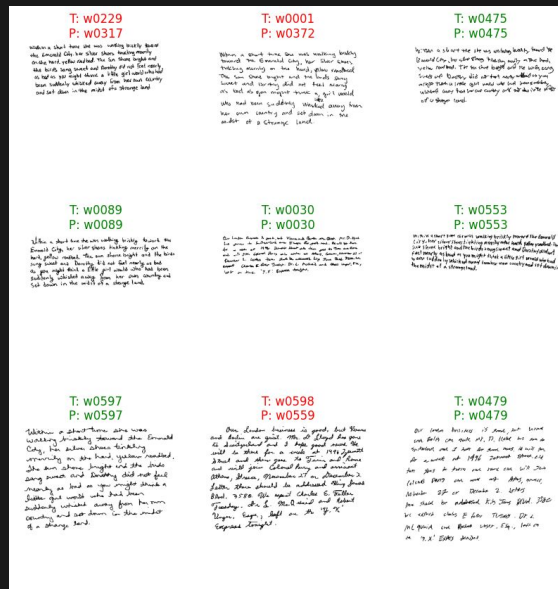
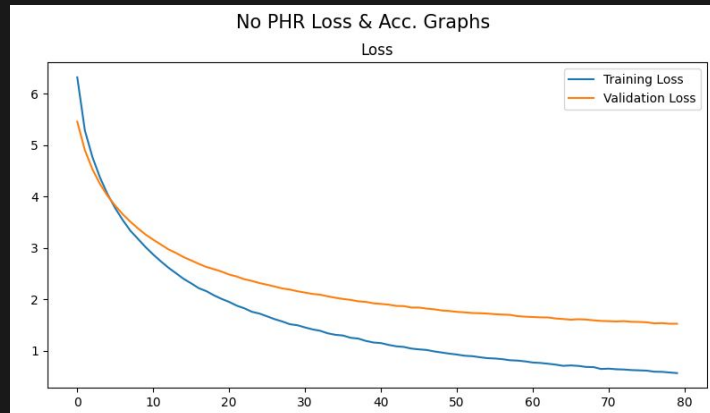
- Removed "PHR"
- 10 – 3 – 5 split

Train Accuracy	91%
Val. Top-3 Acc.	84%
Test Accuracy	67%
Test Top-3 Acc.	84%

LIME: Top Supportive Features

LIME: Pros (Green) / Cons (Red)
LIME top: w0632 | Model: w0632 (23.1%)

When a short time she was walking briskly toward Emerald City, her silver shoes making merry on the hard, yellow roadbed. The girl **longer brighter** and the birds **long** screeled and Dorothy **did not feel** slowing as fast as **yellow** from her **and** **scream** and set down in the midst of a **Dorothy** yard.





Questions?

A Computational Approach to Forensics Handwriting Analysis

Zainab Siddiqui
Data Capstone II





CSAFE Director



Alicia L. Carriquiry

She/her/hers

Distinguished Professor and President's Chair
Director of the Center for Statistics and Applications in Forensic
Evidence

I am interested in the principled application of statistics to solve substantive problems in disciplines including forensic science, nutrition epidemiology, plant and animal breeding, and others. I gravitate toward the Bayesian framework, but I am not obsessive.

I have had the good fortune of contributing to the profession through a series of leadership and editorial roles and have been generously recognized by multiple organizations.

Education

PhD, Iowa State University, Ames, IA, 1989-1985

Affiliations

Director, CSAFE →

Problem: Handwriting Analysis is...

1

Time consuming

2

Subjective

3

Dependent on
expert labor

Why It Matters

1

Backlogging/
delaying
investigations

2

Limited access to
experts

3

Incorrect
decisions have
dire consequences



Stakeholders

- **Forensic Labs:** access to necessary tools
 - Government Labs also have a significant need for questioned document analysis
- **Investigators/Law Enforcement:** faster results, more information
- **Other Computational Forensic Scientists:** collaboration on further development

Model Overview

We are able to filter our suspects down to 3 individuals with 73% accuracy.

84%

When trimming data exclude shorter prompts.

475

Writers/individuals to “choose” from, making the randomness rate 0.2%.

12,500+

Images in dataset.

Model Pipeline

Acquire questioned document

01

Take high quality, black & white image scan

02

Apply initial (minimal) preprocessing and feed into model.

03

Attain results, supporting investigation and further conclusions to support decision making.

04

Expected Impact

Support Decision Making

A tool for filtering;
requires human
verification.

Expedite Investigations

Can be used at
any stage of
investigation.
More cases get
processed—
and faster.

Reproducible and Quantifiable

More consistency,
as well as
referenceable
confidence/
similarity scores.

Integration with Database

Like with CODIS &
AFIS, can be used
naturally along
with reference
information.

What Data Can & Cannot Tell Us

Key Insight:

Handwriting exhibits stylistic characteristics that can be picked up on by AI models (i.e., CNNs).

Key Insight:

Models can significantly narrow field of search and aid in decision-making.

Key Limitation:

In current form, is a "closed set" model; cannot be applied to new situations.

Risks & Consequences

False Negatives:

Can potentially lead investigators away from the true lead and down rabbit holes.

False Positives:

Worst case scenario; contributes to incorrect suspicion and building a case against the wrong person.

Prevention:

AI models should not be used as sole decision makers, but rather as a tool that contributes to a piece of the larger puzzle.

Proposed Integration

Context

Tool for forensic analysts
Allows for fast actionable
direction, especially in
situations where time is
a limiting factor.



Key Players

Forensic Scientists

Prosecutors

Investigators

Law Enforcement

Thank you